Availability-Aware Multi-Signal Fusion and the Operating Envelope of a Machine-Generated Media Detector

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***Abstract*—Fully synthetic images and video are now produced by thousands of publicly available generators, and a detector relying on any single cue fails whenever that cue is absent. We present VidTrust AI, which fuses a pretrained neural classifier, provenance metadata (EXIF/XMP/C2PA) and a fast Fourier transform high-frequency energy ratio under an *availability-aware* rule: a signal that cannot run is removed from the weighted average and the remaining weights renormalised, rather than scored as zero. On a 400-image balanced subset of Community Forensics spanning 50 generators the fused system reaches ROC-AUC 0.9196, rising to 0.9487 under a centre-crop control that equalises processing history without interpolation, the whole gain being 31 fewer false positives on real photographs with recall unchanged to four decimal places. We then measure the operating envelope over eleven degradations applied on top of that control, and report that AUC is nearly blind to the failure that matters: between clean input and JPEG quality 30 the fused AUC moves 0.013, while generated images actually labelled `AI_GENERATED` fall from 172 to 95 out of 200. Mild recompression silently converts detections into abstentions at a frozen decision band. Repeating three conditions *without* the resolution control inverts the sign of the measured robustness (ΔAUC -0.110 controlled against $+0.029$ uncontrolled under JPEG 50), so a naive protocol would report that compression improves the detector; and the frequency signal proves almost exactly compression-invariant, evidence against its being a synthesis cue. We argue that per-signal logging, ablation, and a degradation sweep reported at the deployed threshold rather than by AUC alone should form a minimum reporting standard for fused detectors.**

***Index Terms*—synthetic media detection, multi-signal fusion, robustness evaluation, content provenance, evaluation methodology, model calibration**

I. Introduction

The manipulated media that most needs checking is no longer a face swapped into an otherwise genuine photograph. It is media that is synthetic in its entirety: an image or clip that no camera ever recorded. Latent diffusion models [1] and their thousands of community fine-tunes have made such media cheap, and the detection problem has shifted with it. A face-swap detector [2] needs an original to compare against; a fully synthetic image has no original, and frequently no photographed face at all.

Detectors for this problem are commonly single-model classifiers [3]–[5], which have exactly one way to be wrong and nothing to check it against. An ensemble of independent signals can cover that gap, but only if it handles the case that motivates it. Provenance metadata is the strongest evidence of synthetic origin when present, and is absent from most media in circulation, because social platforms re-encode on upload and strip it. A fusion rule that scores a missing signal as 0.0 treats "no evidence" as "evidence of authenticity" and pushes every stripped file toward a verdict of real.

A second gap is reporting. Detectors are overwhelmingly scored by AUC, which is threshold-free, but a deployed detector decides at a fixed band, and what a user experiences is how often it still says "generated" on generated input. Section VI shows the two diverging widely under ordinary recompression.

This paper contributes (i) an availability-aware fusion rule (Section III-A) that excludes an inoperative signal and renormalises the remaining weights, mapping total failure to maximum ignorance rather than to either verdict; (ii) a two-track protocol with a resolution-normalisation control (Section IV) showing that the standard public set measures resampling history as well as synthesis; (iii) an operating-envelope measurement over eleven degradations (Section VI), reported as ΔAUC and as the verdict shift at the frozen deployed band, with a controlled demonstration that omitting the resolution control inverts the conclusion; and (iv) three negative results invisible to unit tests and aggregate metrics, surfaced only by per-file, per-signal logging: one spectral (Section VI-C) and two operational (Section VII).

The design is deliberately free of fitting: the classifier is a frozen, off-the-shelf model, and the fusion weights and thresholds were fixed by hand before any measurement, so no figure on the public set is optimised against that set. That is what lets the evaluation separate detection from dataset artefacts; Section VI-D reports a fitted fusion that scores higher

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and why we do not adopt it. Because the public set's confounds are characterised rather than removed, its absolute figures are an upper bound on field performance.

II. Related Work

Learned detectors. Wang et al. [3] showed that a classifier trained on one CNN generator transfers surprisingly well across architectures, the single-classifier baseline most later work follows. Corvi et al. [5] showed that detectors trained on GAN imagery degrade sharply on diffusion output. Ojha et al. [4] and Cozzolino et al. [6] responded with detectors on pretrained vision-language features, the latter generalising from a handful of images of one generator, and Tan et al. [7] traced generalisable detection to the local pixel correlations up-sampling leaves behind. Our classifier is an off-the-shelf instance of this line; Section V-A shows why the instance matters more than its architecture.

Robustness and dataset bias. Gragnaniello et al. [8] found reported robustness substantially overstated once compression and resizing are applied. Grommelt et al. [9] showed that common detection datasets, GenImage [10] among them, carry biases in JPEG compression and image size that detectors learn, and that removing those biases substantially improves robustness to compression and cross-generator accuracy. Rajan et al. [11] instead build generated training images by reconstructing real ones through a latent diffusion autoencoder, so the classes are aligned in everything but the decoder's artefacts. Both act on the training data; Section VI is the evaluation-side counterpart, holding processing history fixed at test time and measuring what an unaligned protocol does to robustness conclusions.

Frequency cues and provenance. Up-sampling leaves periodic spectral artefacts, exploited by Durall et al. [12] and Frank et al. [13]; such cues are strong on the generators they were characterised on and fragile under compression and rescaling. The C2PA specification [14] defines signed credentials that travel inside the file. Provenance is high-precision and low-recall by construction and is usually treated apart from detection; combining the two raises the availability problem that motivates this paper. Community Forensics [15] supplies per-image generator attribution, making per-generator failure analysis possible.

III. System Design

Fig. 1 gives the pipeline. A single HTTP endpoint accepts an image or a video and returns a verdict, a fused confidence, and the full per-signal breakdown that produced it.

Classifier ($w = 0.60$). A pretrained image classifier scores the image, or every sampled frame of a video, and its label distribution is collapsed into $P(\text{machine-generated})$. Generic labels such as `LABEL_0`/`LABEL_1` are *not* mapped by guessing an index, since a wrong guess silently inverts the signal; an unrecognised vocabulary makes the signal report unavailable. The model throughout is a SwinV2 [16] classifier (`haywoodsloan/ai-image-detector-deploy`, 744 MB) selected by measurement in Section V-A.

Provenance ($w = 0.25$). A generator fingerprint (a known generator name in decoded EXIF or XMP text, or a C2PA/JUMBF manifest marker) yields 1.0; genuine camera EXIF (`Make` and `Model` present, no generator string) yields 0.0. A file carrying neither makes the signal report *unavailable*, the case Section III-A exists to handle.

Frequency ($w = 0.15$). The image is converted to greyscale, resized to 512×512, and its DC component removed. The fraction of total 2-D FFT energy outside a central low-frequency disc of radius $0.03125 \times 512 = 16$ px is then mapped linearly onto $[0, 1]$ over $[0.05, 0.45]$. **Both endpoints are unfitted**: they bracket the range over which the ratio moves at all, and encode no measured boundary between generated and real.

The three consult independent *evidence*, a neural judgement about pixels, a metadata lookup and a spectral statistic, so a failure mode of one is not automatically a failure mode of another.

A. Availability-Aware Fusion

Let $S = \{\text{model}, \text{metadata}, \text{frequency}\}$ with fixed weights w_k, and let $A \subseteq S$ be the subset of signals that successfully produced a score. The fused confidence is

$$c = \frac{\sum_{k \in A} s_k w_k}{\sum_{k \in A} w_k}, \qquad c = 0.5 \quad \text{if } A = \varnothing. \tag{1}$$

Equation (1) renormalises over the available signals rather than zeroing missing ones, and this is the central design decision. A score of 0.0 means "this looks real"; unavailability means "no evidence either way". Conflating them drifts every metadata-stripped file toward a verdict of authentic, and stripping metadata is exactly what social platforms do on upload. With provenance absent, the classifier and frequency weights rescale from $0.60/0.15$ to $0.80/0.20$. The $A = \varnothing$ case maps to 0.5, maximum ignorance, which falls inside the abstention band, so a fully degraded system declines to answer rather than reporting "real"; likewise a failed classifier is not an error response, since (1) renormalises and the other two signals still answer.

The verdict is a three-way thresholding of c: $c \geq 0.65 \Rightarrow$ `AI_GENERATED`; $c \leq 0.35 \Rightarrow$ `LIKELY_REAL`; otherwise `UNCERTAIN`, which is a first-class outcome and is counted separately from correct and incorrect throughout rather than forced into a confusion matrix.

IV. Evaluation Methodology

Track A (public). 400 images from Community Forensics [15] (CC BY-NC-SA 4.0): 200 generated, spanning 50 distinct generators at 4 images each, all of the `LatDiff` family, and 200 real from FFHQ [17].

Track B (hand-collected). 16 files: 13 labelled (12 images and one video; 4 generated, 9 real) and 3 unlabelled plumbing fixtures excluded from every count. The labelled files are camera originals with intact EXIF, images exported from commercial generators, and files deliberately routed through a messaging platform so their metadata is stripped in transit.

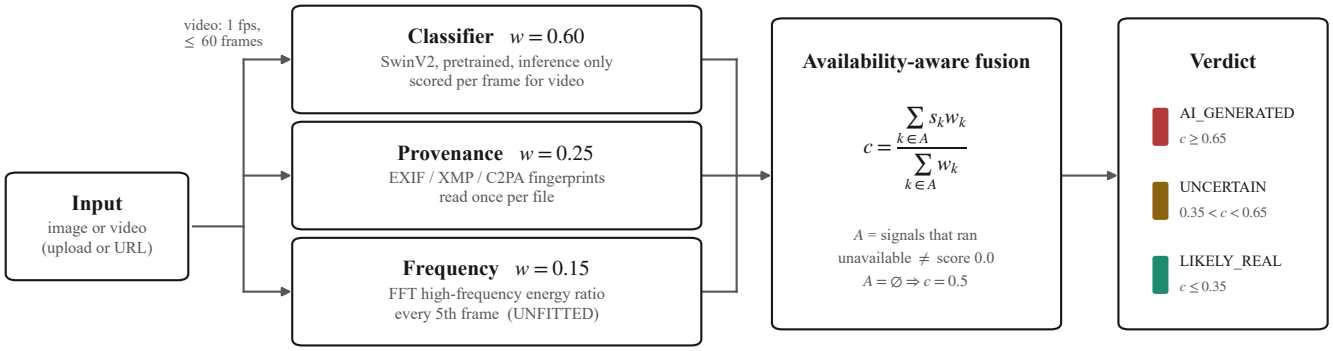


Fig. 1. Pipeline. Three independent signals are computed, then fused over only those that actually ran. Video is treated as images over time: frames are sampled at 1 fps, or spread evenly over longer clips, to a 60-frame cap, and the classifier scores each one.

The tracks are never averaged. Public datasets are redistributed re-encoded, and **0 of the 400 Track A images carry any EXIF**, so averaging would let 400 images with a structurally dead provenance signal drown 13 that exercise it. Track B's 12 labelled images also selected the classifier (Section V-A), so its results are a functional check, never an accuracy estimate.

Metrics. Accuracy, precision, recall and F_1 are computed over *decided* files only, with abstentions excluded and reported separately as coverage: a detector that abstains on half the set and is right on the rest is not accurate. ROC-AUC is computed over all labelled files including abstentions, since AUC ranks a continuous score and never consults a threshold; it is obtained through the rank (Mann–Whitney U) identity with ties averaged.

A. Resolution Normalisation

Every Track A image is analysed as distributed (*naive*) and after a control we call *resolution normalisation*. Real images in this set are natively 1024×1024 and generated ones 512×512, with no overlap. Every detector resizes to 512 internally, so naively the real images arrive downsampled, their high-frequency content destroyed, while the generated ones arrive untouched, and a frequency statistic can separate the classes by measuring who was downsampled rather than who was synthesised.

Resizing everything to a common size does *not* fix this: a 1024^2 image is still downsampled $2\times$ and a native 512^2 one is not, so the operation's effect depends on exactly the variable that differs by class. We instead take a **centre crop of** 512×512 **native pixels**, written losslessly as PNG; nothing is interpolated, so processing history is identical across classes. The crop does leave framing unequal, a sub-region of a large image against the whole of a small one, which is a content difference and reported as such. Normalisation is refused on Track B, since re-saving discards the EXIF that track exists to test.

TABLE I
CANDIDATE CLASSIFIERS ON THE 12 LABELLED HAND-COLLECTED IMAGES. INFERENCE ONLY; NO FINE-TUNING.

Model (HuggingFace identifier)	Correct	Mean, gen.	Mean, real
`haywoodsloan/ai-image-detector-deploy`	12/12	0.987	0.002
`Ateeqq/ai-vs-human-image-detector`	11/12	0.752	0.001
`dima806/ai_vs_human_generated_image_detection`	8/12	0.012	0.011
`Organika/sdxl-detector`	5/12	0.438	0.574
`dima806/ai_vs_real_image_detection`	4/12	0.766	0.694

V. RESULTS

A. Classifier Selection Is a Measurement, Not a Model-Card Decision

The classifier was initially an SDXL-specific detector chosen from its model card. Measured on the 12 labelled hand-collected images it was not merely weak but *anti-correlated*: mean score 0.438 on generated images against 0.574 on real photographs, scoring genuine iPhone originals at 0.998 "artificial" and a commercially generated image at 0.003. No weight and no threshold can repair a signal that points the wrong way. Five off-the-shelf candidates were then scored over the identical files (Table I), and the separation between best and worst is far larger than any fusion parameter could compensate for: for a fused detector, the choice of the heaviest component dominates the choice of how to combine it.

Track B functional check. The provenance branch behaves as designed on all 13 labelled files. The four generated images score 1.0, three on a C2PA manifest and one on a generator signature; the four camera originals carry genuine EXIF and score 0.0; and the five files routed through the messaging platform, four images and the video, report provenance unavailable, so (1) renormalises over the other two signals. All 13 fused verdicts are correct and none abstains: a check of

TABLE II

TRACK A, $n = 400$; METRICS OVER DECIDED FILES. LOWER BLOCK: ABLATION FROM THE PER-SIGNAL DUMP, OMITTING "ENSEMBLE − PROVENANCE" AND "− FREQUENCY", WHICH EQUAL THE FULL ENSEMBLE AND CLASSIFIER-ONLY WHILE PROVENANCE IS DEAD.

	naive	normalised
Accuracy	0.7595	**0.8384**
Precision	0.7107	**0.8152**
Recall	0.8731	0.8731
F_1	0.7836	**0.8431**
Coverage	0.9875	0.9900
TP/FP/TN/FN	172/70/128/25	172/**39**/160/25
ROC-AUC, full ensemble	**0.9196**	**0.9487**
classifier only	0.8397	0.8947
frequency only	0.9012	0.9129
provenance only	0.5000	0.5000

every branch and the renormalisation path on real files, not an accuracy estimate.

B. Track A Headline, the Control, and the Ablation

Table II gives the aggregate result. **Recall is identical in both conditions to four decimal places**, by construction: the crop is a no-op on the natively 512^2 generated images, whose scores are byte-identical across runs. The entire normalised gain is therefore 31 fewer false positives on real photographs (70 to 39), with real images scoring above 0.5 falling from 72/200 to 40/200, exactly what removing a resampling confound that inflated the real side should do. The characteristic error is calling a real portrait generated.

The lower block re-fuses the dumped per-signal scores without re-running inference; dropping a signal is arithmetically the same as its being unavailable at runtime, so the ablation exercises the real fusion path. **The provenance row is not a measurement**: every image abstains, and AUC 0.5000 is the definition of a constant predictor. **The frequency signal is lopsided**, precision 1.0000 at recall 0.3237 in both conditions: it never fires falsely here and catches about a third of generated images, the profile of a signal keyed to a visually distinct subset.

VI. THE OPERATING ENVELOPE

A detection score is only meaningful inside a stated operating envelope, and we measured ours. Eleven degradations are applied to every Track A image *after* the centre crop of Section IV-A, so both classes enter each condition with identical processing history, and each condition re-evaluates all 400 images, 4800 inferences in total (Table III, Fig. 2).

A. AUC Is Nearly Blind to the Failure That Matters

Between clean input and JPEG quality 30, the fused AUC moves from 0.9487 to 0.9355, a change of 0.013 that a reader would describe as robust. Over the same interval the generated images the system actually labels `AI_GENERATED` fall from 172 to 95 out of 200, a 45% loss of detections. The mechanism is visible in the per-signal dump: the mean classifier score on generated images falls from 0.862 to 0.485, below the 0.65 threshold, while the mean on real images falls further still, 0.225 to 0.046. The *ordering* is preserved, which is all AUC measures, but the generated distribution has slid out of the `AI_GENERATED` band into `UNCERTAIN`. Precision rises from 0.8152 to 0.9500: the degraded system is not making more mistakes, it is answering a different, much easier question and declining the rest.

TABLE III

OPERATING ENVELOPE, TRACK A, $n = 400$ PER CONDITION, EVERY CONDITION APPLIED AFTER THE CENTRE CROP SO BOTH CLASSES DEGRADE IDENTICALLY. "DETECTED" IS GENERATED IMAGES LABELLED `AI_GENERATED` AT THE DEPLOYED BAND, OUT OF 200. PROVENANCE IS UNAVAILABLE THROUGHOUT.

	ROC-AUC			at frozen band	
Condition	Ens.	Clf.	Freq.	Acc.	Detected
clean (crop only)	0.9487	0.8947	0.9129	0.8384	172
JPEG q95	0.9097	0.8177	0.9128	0.7727	149
JPEG q75	0.9314	0.7950	0.9132	0.7744	129
JPEG q50	0.9328	0.7845	0.9119	0.7456	110
JPEG q30	0.9355	0.7683	0.9098	0.7286	**95**
downscale $\times 0.75$	0.9480	0.8956	0.9068	0.8246	176
downscale $\times 0.50$	0.9306	0.8681	0.9016	0.7679	179
crop to 256^2	0.9157	0.8488	0.9117	0.7854	150
noise σ=2	0.9219	0.8437	0.9117	0.7096	182
noise σ=5	0.9107	0.8159	0.9046	0.7683	132
noise σ=10	0.9344	0.7446	0.8818	0.6892	**75**
messaging app	0.9623	0.8588	0.9068	0.7980	122

Mild recompression is not an attack; it is what happens to every image passing through a messaging application. That realistic composite (long edge resampled to 1080 in either direction, then JPEG 70) lands at 122 detections of 200 with AUC 0.9623, *higher* than clean. Under noise σ=10 only 75 detections remain. Light degradation fails the other way, into false alarms: noise σ=2 lifts the mean classifier score on real images from 0.225 to 0.511 and labels 97 of 200 real photographs generated, against 39 clean, and halving the resolution flags 71.

B. Without the Control, the Conclusion Inverts

Three conditions were also run on the *uncontrolled* set, degrading the images as distributed with their native resolutions intact; this arm is a second result, not a mistake to be avoided. Under JPEG 50 the classifier's AUC change is -0.110 with the control and $+0.029$ without; under the messaging condition, -0.036 with and $+0.004$ without. Both signs flip, so a protocol degrading images as they come would report that compression and platform re-encoding *improve* this detector. Under noise σ=10 the sign agrees but the magnitude is overstated twofold, -0.150 against -0.300. The mechanism is the asymmetry of Section IV-A: noise on a 1024^2 real image is partly averaged away by the internal resize to 512, while on a native 512^2 generated image it is not, so the degradation carries label information.

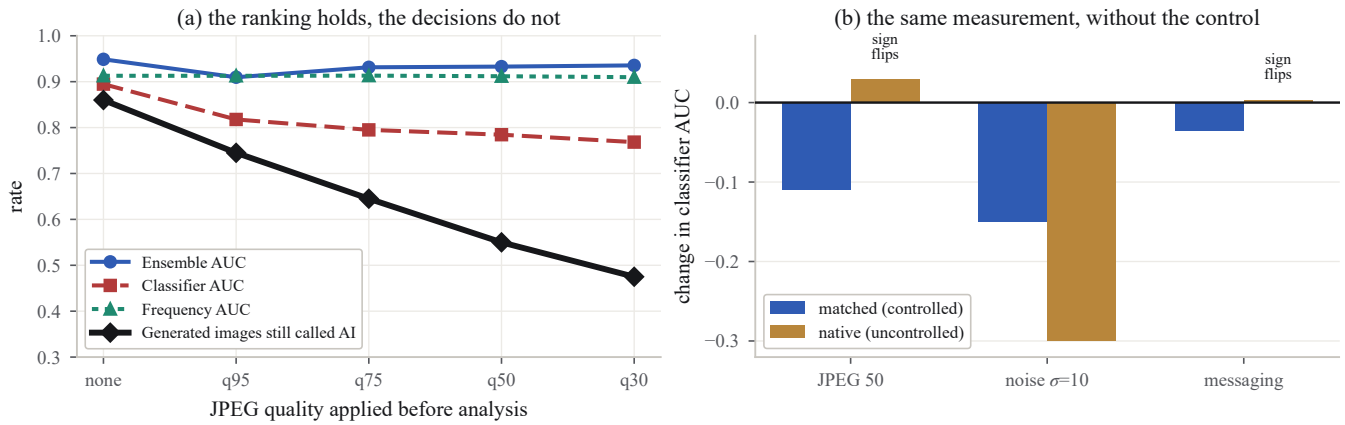


Fig. 2. (a) Under JPEG recompression the ranking quality of all three signals is flat while detections fall by 45%: AUC does not see the failure. (b) Classifier ΔAUC with and without the resolution control; two of three conditions change sign once the control is removed.

C. Compression Invariance of the Frequency Cue Points to Content Confounding

Across all four JPEG levels the frequency signal's AUC spans 0.9098 to 0.9132, a range of 0.0034 against a clean value of 0.9129. We had predicted the opposite. If the signal measured the spectral trace of generative up-sampling [12], [13], JPEG should erase it, since quantising high-frequency DCT coefficients is precisely what JPEG does and precisely where that trace lives; a statistic that survives quantisation of the band it claims to read is not reading that band's artefacts. The economical explanation is gross image structure, aligned FFHQ portraits being smooth where the generated half is detailed illustration, which JPEG preserves. The signal stays labelled unfitted; if a content-matched set confirms this, setting its weight to zero is the correct outcome.

D. The Confidence Is Not a Probability

The fused c is a weighted average of three incommensurate scores, and nothing makes it a calibrated posterior; measured, it is not one. Expected calibration error is 0.1154 on the held-out half, and the empirical rate is not even monotone in c: the 0.7–0.8 bin holds 32 files of which only 31.2% are generated, the 0.1–0.2 bin 28 of which 46.4% are. Platt scaling fitted on the selection half makes it slightly worse (ECE 0.1363, Brier 0.1241 against 0.1187 raw), as expected when the empirical rate is non-monotone rather than merely miscaled, so c is a ranking score, not a probability to print as a percentage. Fitting the fusion itself, by logistic regression over the two available per-signal scores under leave-one-generator-out cross-validation with 50 folds, reaches out-of-fold AUC 0.9674 against 0.9487, with implied relative weights 0.26 model and 0.74 frequency. We do not adopt it: reweighting toward the signal the preceding subsection argues is measuring content fits the confound, not synthesis.

VII. Two Additional Negative Results From Per-Signal Logging

Both surfaced only because the harness logs every signal's score *and* availability flag per file, not just the fused verdict.

A three-character string inverted the provenance signal. The generator signature list contained `"veo"`, and in a few hundred kilobytes of compressed image data a three-character case-insensitive byte sequence occurs by chance with high probability. Provenance consequently reported "generator signature found" at full confidence on **62 of 200 real photographs** (31.0%), firing across the set on 65 real images against 23 generated ones: the signal meant to prove synthetic origin fired more often on real photographs than on generated ones, each time adding a quarter of the fused weight toward `AI_GENERATED`, while every unit-level behaviour remained correct. The fix separates concerns rather than deleting: the full list is matched in decoded EXIF and XMP text, where a short generator name is a word, and a strict list of six or more characters plus structural C2PA markers in raw binary. Verified afterwards on the same 200 real images: 0 false positives.

The ensemble once ranked below its lowest-weighted component. With the original classifier, the fused ensemble reached AUC 0.7918 naive and 0.8461 normalised while frequency alone, at the smallest weight of 0.15, reached 0.9012 and 0.9129. We reported that rather than reweighting, which on this dataset would have fitted the content confound. After the classifier replacement the ensemble dominates every constituent (Table II), yet the frequency signal is the same code on the same files with the same AUC to four decimal places, and the fusion rule and weights are unchanged: only the classifier moved. An ensemble is not robust to a bad heaviest component, and an ablation table is what makes that visible. The error structure moved too. The original classifier never caught six generators and always caught eight, the signature of a specialist recognising SDXL-family output; the replacement misses no generator on all four of its images and slips at worst 2 of 4, near the decision boundary rather than at zero.

VIII. Limitations

Dataset confounds. Three properties of Track A inflate its numbers. (i) Zero EXIF across all 400 images, so provenance is structurally dead and every figure in Sections V and VI is a two-signal result. (ii) Every real image is an FFHQ face while the generated half is varied digital art and photography, so the classes are partly separable on subject matter, which the centre crop plausibly intensifies. (iii) The resolution asymmetry of Section IV-A. Together these make Track A's figures an upper bound, and are why no fitted parameter here is adopted.

Unfitted weights. $0.60/0.25/0.15$ were chosen by inspection before any measurement; that they now beat any single signal is evidence about the classifier, not the weights.

Scope. Real C2PA manifest parsing is not implemented, so the system reports that a manifest is present but never who signed it. Track A is images only, so the frame-sampling path is exercised by one labelled clip and by plumbing fixtures, and every claim here is an image claim. On an unaccelerated CPU-only baseline (4-core Intel i5-1135G7), in-process inference with the 744 MB SwinV2 takes 1.29 s per Track A image on average, and a full-resolution camera original takes 4–6 s end to end through the API.

IX. Future Scope

A matched evaluation set, real and generated images sharing subject matter, resolution and processing history, would remove the confounds of Section VIII: the prerequisite for attributing the frequency signal's discrimination, fitting the weights, and recalibrating c. **Video** needs a labelled set of clips from current text-to-video generators, on which temporal cues such as the dispersion of per-frame scores can be measured. **Provenance** needs full C2PA parsing with signature verification, and media fetched from platform links, re-encoded and stripped, needs its own envelope, as does adversarial post-processing.

Per-input reliability. Because the envelope is measured, a detector can estimate an input's condition, such as JPEG quality from its quantisation tables, and report the detection rate measured under the nearest condition beside its verdict; ours does, though its value to users is untested. **Faithful attribution.** Against occlusion on 20 images, Grad-CAM on this classifier agrees at median Spearman $\rho = -0.02$, no better than chance, so the system draws no saliency map; a method passing that test remains open.

X. Conclusion

Our three-signal detector treats an unavailable signal as absent evidence, not evidence of authenticity, and reaches ROC-AUC 0.9196 on a 400-image public set spanning 50 generators, 0.9487 under a resolution control.

The methodological claims are the ones we would most like carried forward. First, report a degradation sweep at the deployed threshold, not by AUC alone: between clean input and JPEG q30 our AUC moves 0.013 while 45% of detections disappear. Second, hold resolution fixed in that sweep, or its conclusions invert: two of three conditions change the sign of the measured ΔAUC without the control. Third, log per signal: that exposed a provenance signature firing at full confidence on 31% of real photographs while every unit test passed, and an ensemble ranking below its lowest-weighted component. All three are cheap, and belong in the minimum reporting standard for fused detectors. Every figure and table is produced by a script in the project repository, recomputed from one per-file, per-signal dump without re-running inference.

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